Populous for Mac

Player's manual: every button, icon, click and key



A native Swift port of Bullfrog's *Populous* (Amiga, 1989). This manual describes the controls of the port exactly as they are wired in the current build, including the few places where the port differs from the original or adds a Mac convenience. Screenshots were taken from the port itself running with the original game data.

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1. Starting the game

The port contains no game data. It reads the graphics, sounds, worlds and text from your own Populous disk image (an **.adf** file) every time it starts.

Build and open the app from the project folder:

```
./make-app.sh
open Populous.app
```

The first time, a Mac file dialog asks for the disk image. The choice is remembered. To skip the question altogether, keep the .adf file next to Populous.app, or pass its path on the command line:

```
swift run Populous "path/to/Populous.adf"
```

The window

The game draws a 320 x 200 picture, scaled up by the largest whole number that fits the window and centred on black. You can resize the window freely; use **Cmd-F** (Window menu, Enter Full Screen) for the biggest picture. Over the game picture the Mac arrow is hidden and the game draws its own pointer, exactly as the original did. The point that clicks is the **top-left pixel** of the pointer sprite, not its middle.

The Mac title bar is used as the game's message line. During play it shows the world's name, your mana, both populations, the current behaviour and any notices (see section 15).

Mouse buttons

Populous uses both mouse buttons. On a trackpad or a one-button mouse, a right click is a two-finger click or Control-click, whichever your Mac is set to. Throughout this manual **left** means the primary button and **right** means the secondary button.

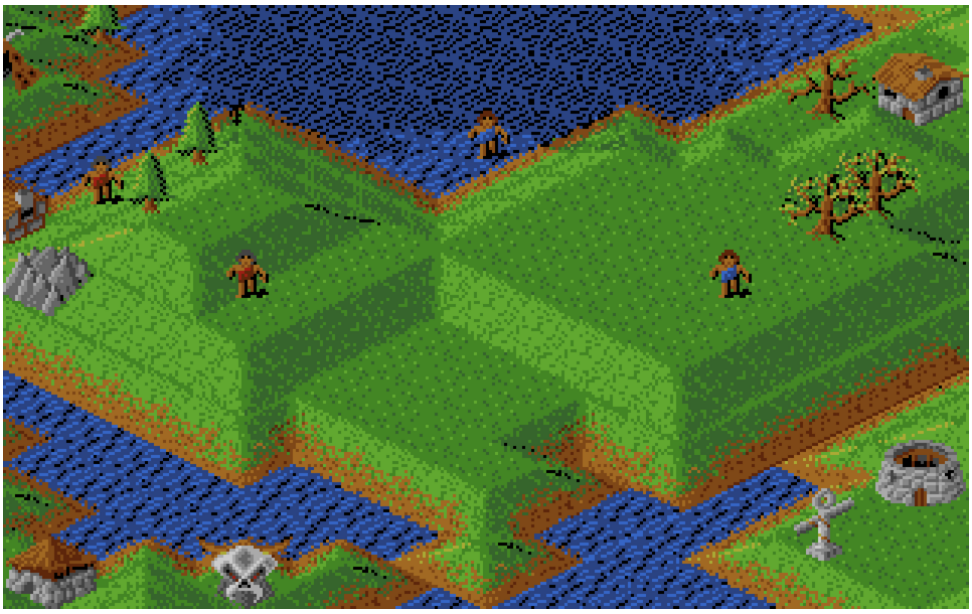
2. The title picture



The three buttons at the lower right of the title picture.

Click on	What happens
TUTORIAL	Starts a practice world: fifteen of your people against three, 10,000 mana to begin with, and a slow harmless opponent. It starts **paused**, as in the original: press **P** or click the zzZ icon to begin. In the tutorial you may reshape land, attack and use earthquakes, swamps and knights, but not volcano, flood or Armageddon.
CONQUEST	Opens the World to Conquer box for the first world, Genesis (section 3). The 495 worlds are played in order.
CUSTOM	Starts a random world at once, with every power allowed to both sides. Only in a custom game can the option boxes (section 12) be changed.
Anywhere else	Starts the animated demo immediately. The demo also starts by itself if you leave the title alone for a few seconds. Any click or key during the demo returns you to the title.

Keys on the title picture: **Cmd-N** is the same as clicking CONQUEST. **Cmd-O** opens a saved game straight away. **Cmd-Q** quits.



The demo: land drops into place, figures act out a scene, credits scroll. Click or press a key to leave it.

3. The World to Conquer box



The box for Genesis, the first world.

The box describes the world you are about to play: its name, its battle number (worlds come in steps of five, so battle 1 is world 5), the landscape (Grass Planes, Desert, Snow and Ice, Rocky), how fast and how well the computer plays, the rules of the land, the swamps and the water, and then for each side the starting population and which of the five big powers it may use.

Button or key	What happens
START GAME (left button) or Return	Begins the world shown. The view opens on your leader.
NEW GAME (right button)	Turns the name line into a typing field so you can go to any world by name (below).
Esc	Closes the box, but only if you opened it yourself during a game with Cmd-N. The box that follows a win cannot be dismissed.

Typing a world's name



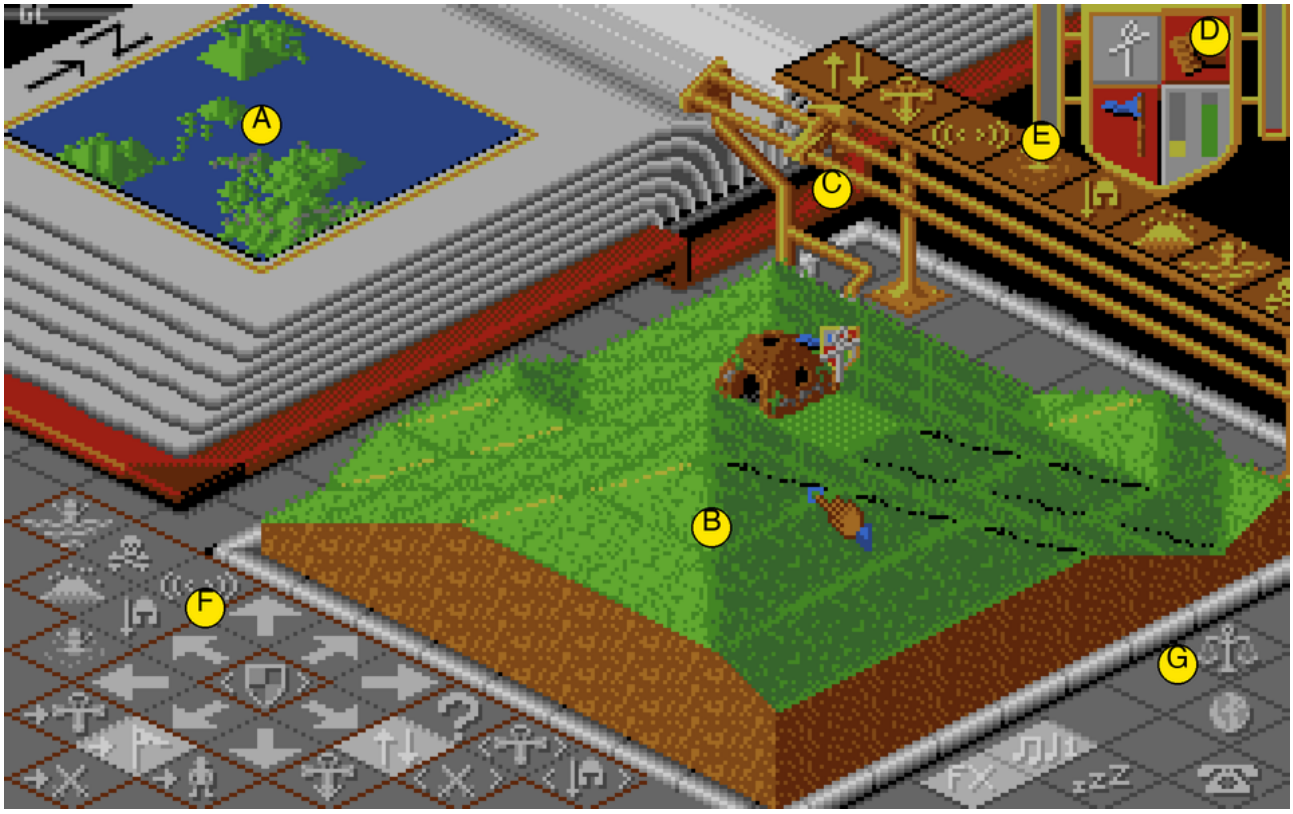
The typing field. The cursor blinks; letters are shown in capitals.

Type up to fifteen characters. Letters are turned into capitals as you type; **Delete** removes the last one. Press **Return** or click START GAME: if the name is a real world, the box fills in with that world and you can start it. If not, the line below says NO SUCH WORLD and you can go on typing. Clicking NEW GAME again clears the field. Every world's name is shown in its own box and in the title bar, so you can note the name of a world you want to return to. The port picks the nearest multiple of five, which is how the original numbers its battles.



An unknown name.

4. The game board at a glance



The board with a person being followed, so the shield is filled in.

	Area	What it is and how to use it
A	The map	The book at the top left shows the whole 64 x 64 world: land colours by height, blinking dots for houses, people and fights. Click or drag on it to move the close-up (section 6).
B	The close-up	An 8 x 8 cell window on the world where you act. Left click raises land, right click lowers it (section 5).
C	The mana rail	The gold rail with eight icons. A pointer slides down it as your mana grows and shows which powers you can afford (section 9).
D	The shield	Details of the person you are following: side, weapon, picture and two bars (section 10). Blank when nobody is followed.
E	Population bars	Two small bars beside the shield: blue on the left is your population, red on the right is the enemy's.
F	The command panel	Divine powers, view arrows, pointer modes, go-to icons and the four behaviours (sections 7 to 9).
G	The side panel	The scales, the globe, music, sound effects, pause and the message icon (section 11).

The icons that are switched on are drawn in brighter stone: the behaviour in force, the pointer mode, music, effects and pause. (This lighting is the port's own; the original marked its icons differently.)

5. The mouse in the close-up view

Raising and lowering land

Your people cannot be commanded directly. What you can do is shape the ground under them. Flat land is everything: a walker who finds flat land settles and builds; the more flat land around a settlement, the bigger it grows, from a tent through huts and houses to a castle, and the more people and mana it makes.

Click in the close-up	Effect	Mana
Left button	Raises the corner nearest the pointer's tip by one step.	10
Right button	Lowers that corner by one step. The right button lowers land in every pointer mode.	10

Each click is an order carried out on the next tick of the game. The original's rules, which the port keeps, refuse the click in these cases, and the port says why in the title bar:

- **Nobody of yours is in view.** You may only reshape land while at least one of your own people is inside the close-up. This is the usual reason a click seems to do nothing: scroll to your people first.
- **The game is paused.** Click zzZ or press P.
- **The world forbids it.** Some conquest worlds cannot be built on, or may only be built up (see the rules in the World to Conquer box).
- **Not enough mana.** Reshaping costs 10 mana, plus a little more when neighbouring corners have to move too.

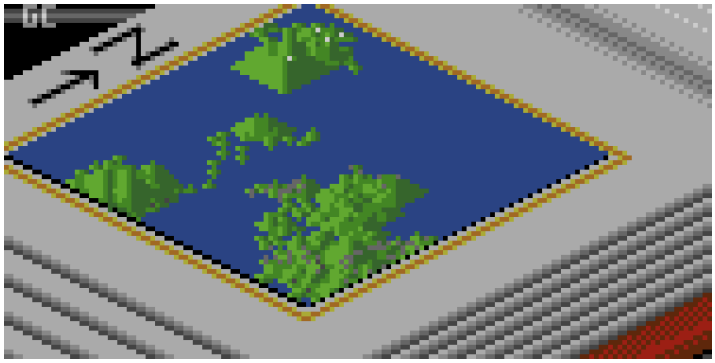
A good opening: flatten the land around your first walkers, let them settle, and keep widening the flat area as new walkers appear. Lower hills into the sea rather than raising valleys where you can; it keeps the land flat for longer.

Pointer modes

The pointer's shape tells you what a left click will do. The three icons at the right of the command panel change it; **Esc** always returns to the hand. The green arrow seen on the title picture and in the boxes before a game begins is a plain pointer with no mode.

	Pointer	Mode	Left click in the close-up does
	Hand	Normal (icon 17 or Esc)	Raises land.
	Shield	Looking (icon 16, the ?)	Follows the person you click on: the shield at the top right then shows their details. Clicking empty ground does nothing except remind you in the title bar that you are looking.
	Arrow with ankh	Placing the papal magnet (icon 18)	Moves your papal magnet to that spot for 200 mana, then returns to the hand. You need a leader for the magnet to be placed.
	Arrow with reeds	Placing a swamp (icon 6)	Scatters swamp around that spot for 5,000 mana, then returns to the hand.

6. The map



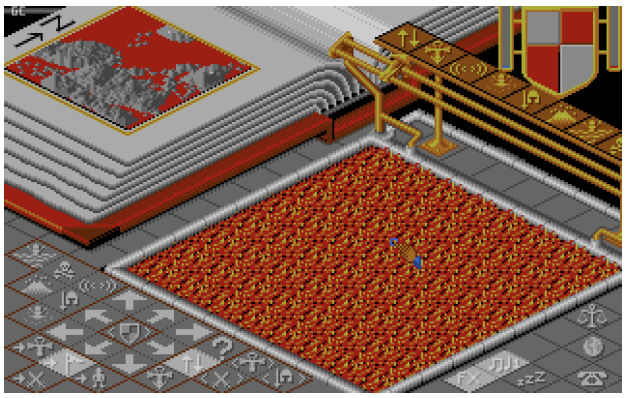
The map after a few minutes of play. Blinking dots mark houses on one frame, walkers and fights on the next.

- **Click** anywhere on the map to move the close-up there. The spot you click lands a little in from the close-up's top corner, as in the original.
- **Drag** with the left button held down to scroll continuously.
- The **arrow keys** and the eight arrow icons on the command panel scroll the close-up one cell at a time (section 7).
- The map shows every side's dots; the close-up only shows what is inside its 8 x 8 cells, so use the map to find the enemy before striking.

Land on the map is coloured by its height and landscape: in Grass Planes the sea is blue and land green; the Desert, Snow and Ice and Rocky landscapes use their own colours, shown below.

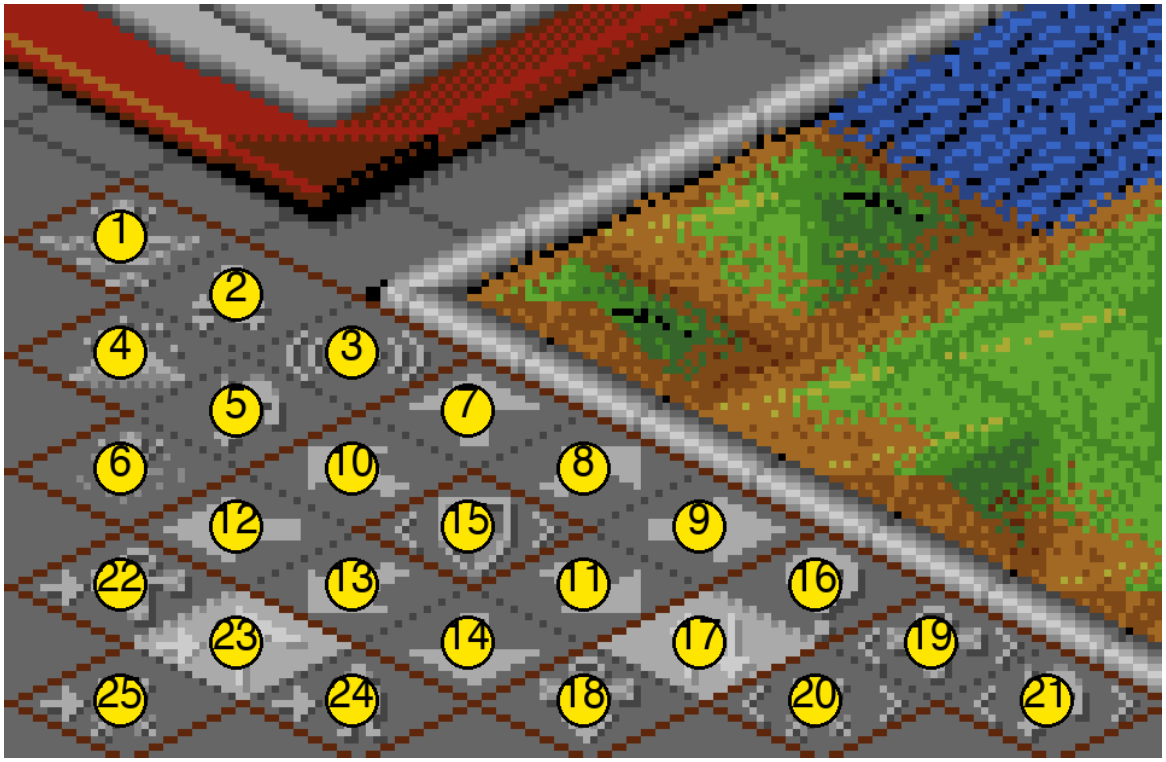


Snow and Ice.



Rocky, with a lava sea.

7. The command panel (lower left)



The command panel, numbered for the table below. Lit icons are the ones switched on.

Divine powers (1 to 6)

Each power needs enough mana and must be allowed on this world (the World to Conquer box and the Options boxes list them). If either is missing the click simply does nothing, as in the original. Section 9 explains the mana rail.

	Icon	Click	What it does	Mana
1	Waves	Left	**Flood.** Lowers the whole world by one step. Everyone standing on the shore drowns, your people included.	40,000
2	Skull	Left	**Armageddon.** Nobody settles any more; both magnets go to the middle of the world and everyone marches there for a final battle. The title bar says ARMAGEDDON while it runs.	80,000
3	Ripples	Left	**Earthquake.** Shakes the land under the close-up, cracking flat ground. Move the view over the enemy first.	2,500
4	Mountain	Left	**Volcano.** Raises a mountain of rock under the close-up. Nothing can be built on rock.	10,000
5	Helmet	Left	**Knight.** Turns your leader into a knight who hunts the enemy and burns their homes. You need a leader (send people to the magnet first).	7,500
6	Reeds	Left, then click in the view	**Swamp.** Changes the pointer; the next left click in the close-up scatters swamp there. Walkers who step in are lost. Press Esc to change your mind.	5,000

View arrows (7 to 15)

The eight arrows round the shield-shaped centre scroll the close-up one cell in the direction of the arrow, exactly like the keyboard's arrow keys: 7 up and left, 8 up, 9 up and right, 10 left, 11 right, 12 down and left, 13 down, 14 down and right. The centre (15) jumps back to the person you are following, if any.

Pointer modes (16 to 18)

	Icon	Click	What it does
16	Question mark	Left	**Look.** The pointer becomes a shield; click a person to follow them (section 10).
17	Up and down arrows	Left	**Reshape.** Back to the hand pointer that raises and lowers land. Same as Esc.
18	Ankh with arrow	Left, then click in the view	**Place the papal magnet.** The next left click in the close-up moves your magnet there (200 mana). Your leader walks to it, and when the behaviour is Go to magnet so does everyone else.

Go-to icons (19 to 21)

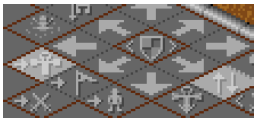
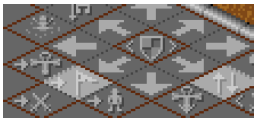
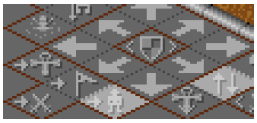
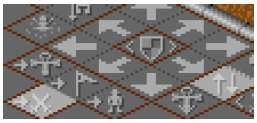
These move the close-up and, except for the magnet, start following the person found, so the shield shows them. Each click moves on to the next one, going round the whole map.

	Icon	Left button	Right button
19	Ankh in brackets	Go to your papal magnet.	Go to your leader and follow them. If you have no leader, go to the magnet.
20	Crossed swords in brackets	Go to the next fight anywhere on the map and follow one of the fighters.	Same as left.
21	House in brackets	Go to your next settlement and follow its occupant.	Go to your next knight.

Behaviours (22 to 25) are described in the next section. The icons half hidden at the left edge of the board are decoration and do nothing.

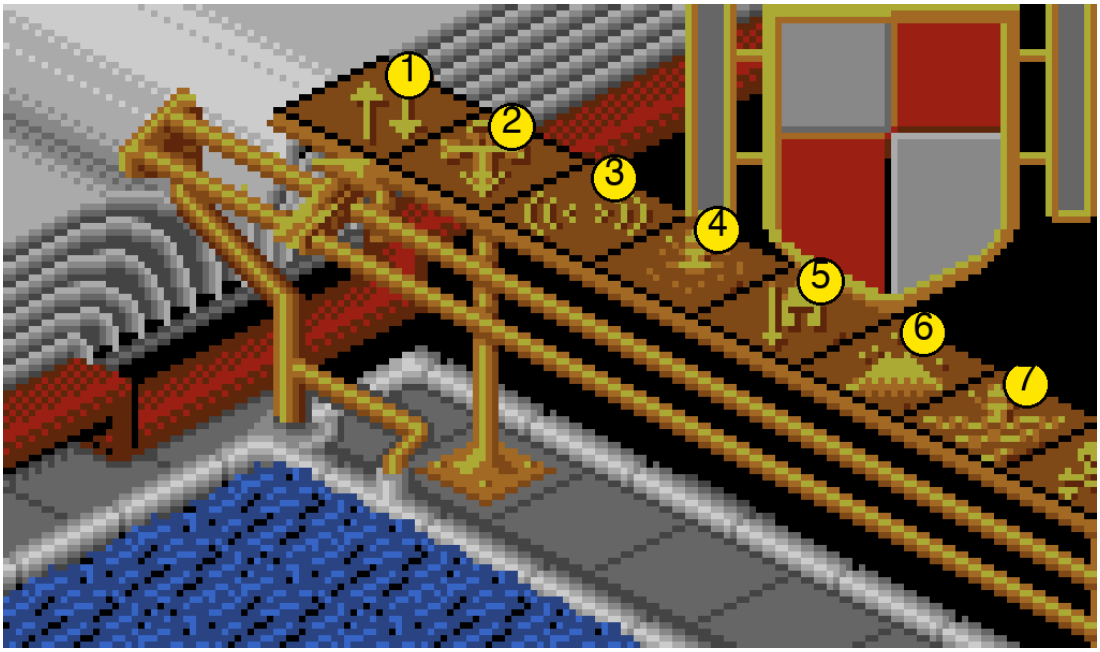
8. Behaviours

The four icons at the bottom of the command panel tell **all** your walkers what to do. Only one is in force at a time; it is drawn in brighter stone. Settled people stay in their houses whatever the behaviour; it is the walkers who obey.

	Icon	Behaviour	What your walkers do
	22	Go to magnet	Walk to your papal magnet, the ankh standing on the land. The first to arrive becomes your leader; the others gather round. Use it to collect people before placing a knight or to pull them out of danger.
	23	Settle	Look for flat land and build on it. This is the normal behaviour and the one most worlds begin with.
	24	Gather	Join together into fewer, stronger walkers. Two walkers who meet merge their strength.
	25	Fight	Seek out the enemy and attack their people and houses.

The title bar always names the behaviour in force. The icons sit in a diamond: Go to magnet on the left, Settle in the middle, Gather on the right and Fight below.

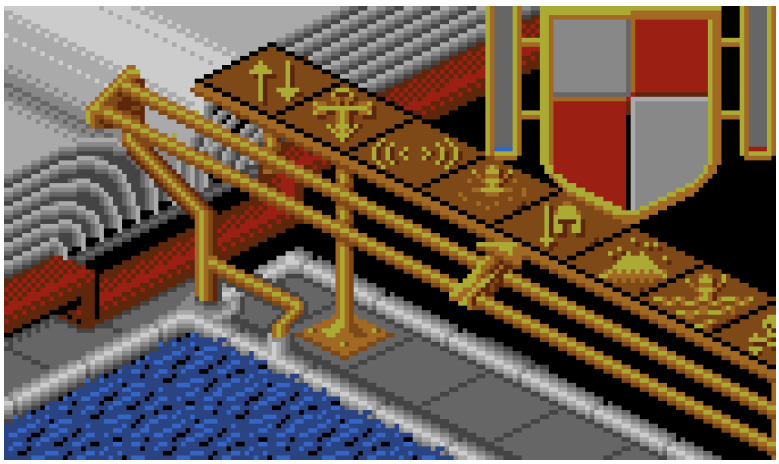
9. Divine powers and the mana rail



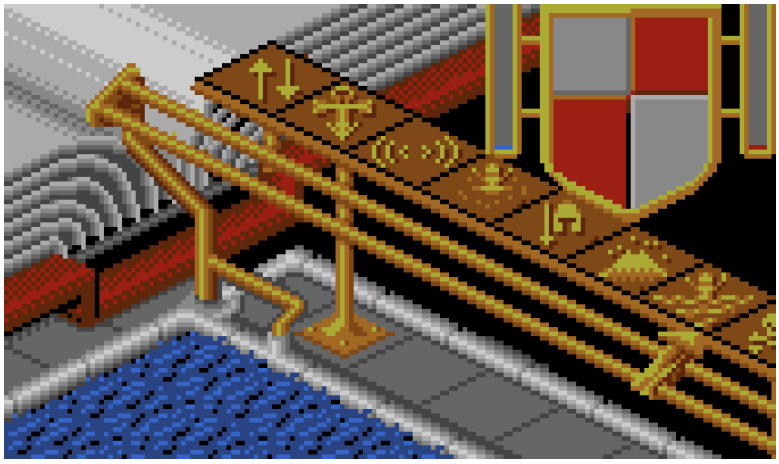
The mana rail at the start of a world. The pointer, a small gold needle, sits at the top of the rail.

Mana pays for everything and grows with your population: every settlement makes a little each tick, bigger ones more. The exact figure is in the title bar. The needle on the rail gives the same information at a glance: it starts at the top left and slides down the rail past one icon for each power you can now afford.

	Rail icon	Power	Mana needed	Where it strikes
1	Up and down arrows	Raise or lower land	10	The corner you click
2	Ankh	Place the papal magnet	200	The cell you click
3	Ripples	Earthquake	2,500	Under the close-up
4	Reeds	Swamp	5,000	The cell you click
5	Helmet	Knight	7,500	Your leader, wherever they are
6	Mountain	Volcano	10,000	Under the close-up
7	Waves	Flood	40,000	The whole world
8	Skull (under the board's edge)	Armageddon	80,000	The whole world



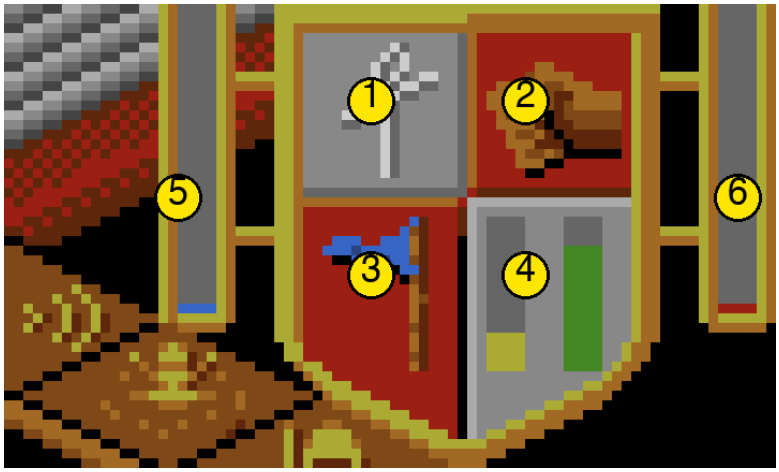
With 9,000 mana the needle has passed the knight.



With 85,000 mana it reaches the bottom of the rail: everything is affordable.

Which powers you may use on a given world is decided by the world (conquest) or by the Options for Good box (custom game). An icon you may not use does nothing when clicked, however much mana you have. Using a power also adds to your score.

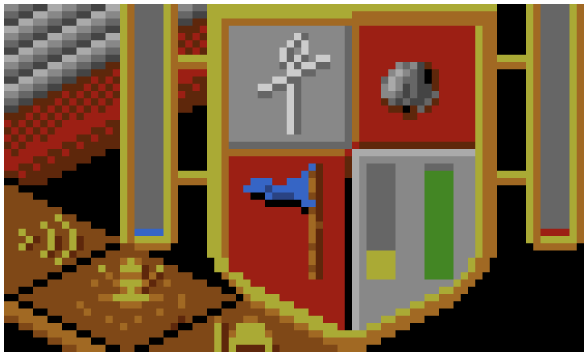
10. The shield and the population bars



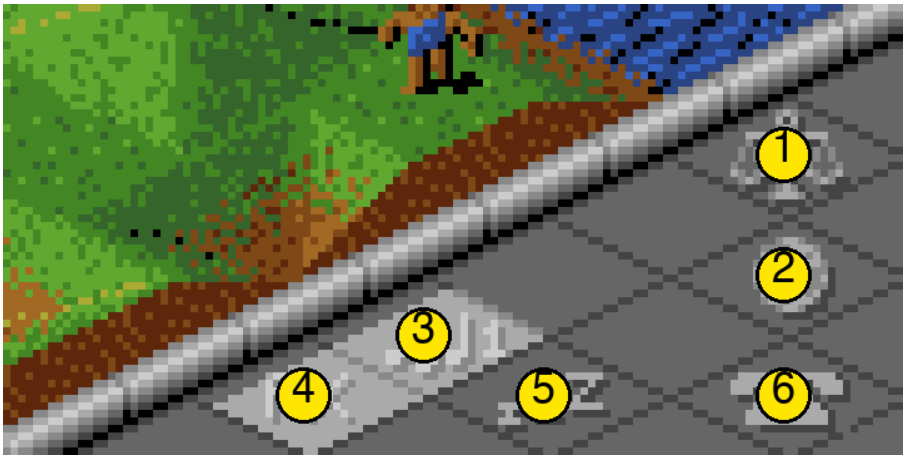
The shield while following your leader, who has settled.

Click the ? icon (16) and then a person in the close-up, or use any of the go-to icons 19 (right button), 20 and 21, and the shield at the top right fills in with that person's details. It stays on them as they move; the centre of the view arrows (15) brings the view back to them. The shield empties when they die or when you follow nobody.

	Quarter	What it shows
1	Top left	The side: an ankh on grey for good (you), a skull for evil.
2	Top right	The weapon, which is the person's rank of strength on this landscape: a bare fist for the weakest, then stone, club, and so on up to the best weapons that only a castle gives.
3	Bottom left	A picture of the person as they are now: walking in some direction, settled (a flag on the house), fighting, or drowning.
4	Bottom right	Two bars. For a walker: their strength in two digits, left bar hundreds and right bar tens, so the strength above is 0 and 10. For a settled person: how good the site is for a house (yellow) and how full the house is (green); when it is full a new walker leaves. For a fighter: each fighter's share of the life left between them, you on the left.
5	Left pillar	Your population, a blue bar growing upwards. The bar is full at 50,000 people.
6	Right pillar	The enemy's population in red.

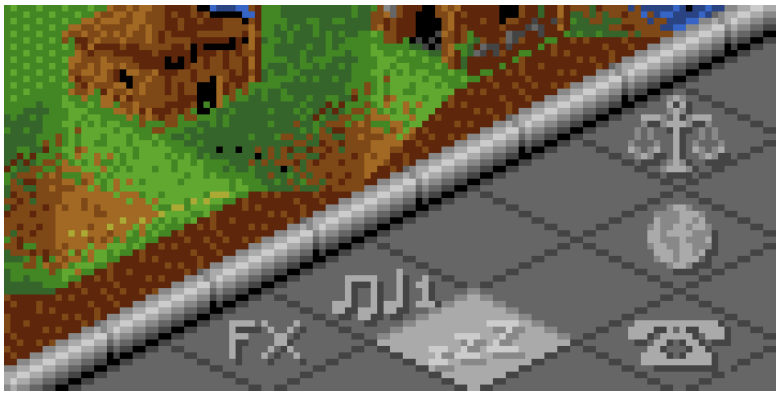

Following a settler: the weapon has gone up to a stone, the house is nearly full.

11. The side panel (lower right)



The side panel. The music, FX and zzZ icons are lit while they are on.

	Icon	Left button	Right button	Key
1	Scales	Opens **Options for Evil**: the computer's abilities, aggression and rate.	Opens **Options for Good**: your own abilities.	
2	Globe	Opens the **Game Setup** box: game type, game options, save, load, next map, restart, surrender.	Same as left.	
3	Musical note	Music on or off.	Same as left.	M
4	FX	Sound effects on or off: fights, fire, the heartbeat and the powers.	Same as left.	S
5	zzZ	Pause. While paused the world stands still, clicks in the view are refused and the title bar says so. Click again to go on.	Same as left.	P
6	Telephone	The message icon of the original. Not ported; does nothing.	Nothing.	



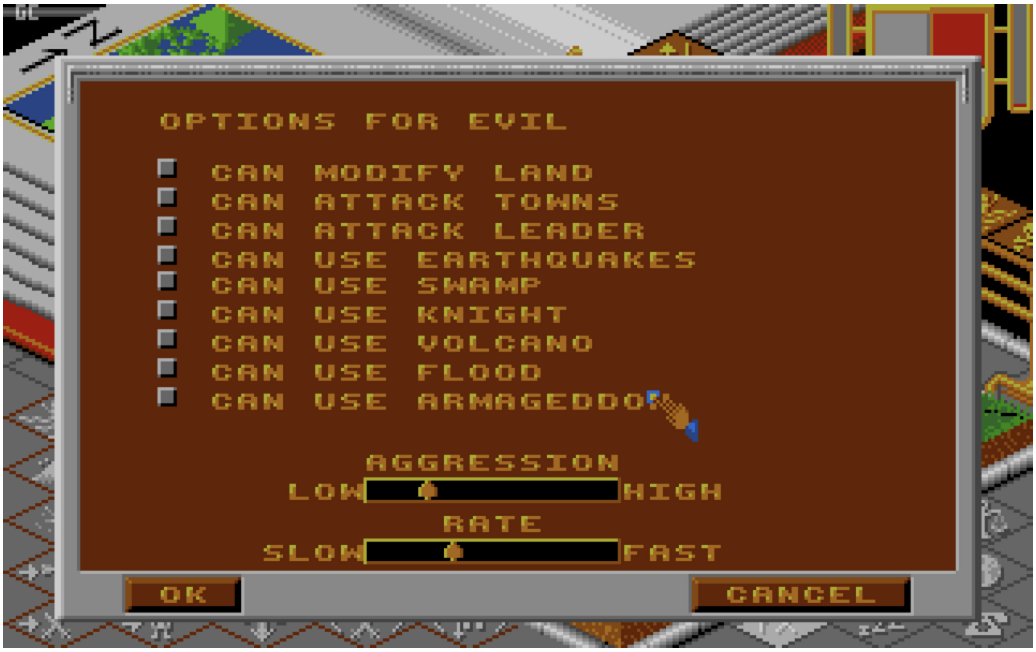
Paused with music and effects off: only zzZ is lit.

12. The option boxes

The three boxes below are the original's, opened from the scales and the globe. While a box is up the game waits. In each box a click on a line toggles it or selects it; **OK** (or Return) keeps the changes and **CANCEL** throws them away. Esc does nothing in these boxes; use CANCEL.

The abilities and the rules can be looked at in any game but only changed in a custom game. In a conquest world a click on them puts "These can only be changed in a custom game" in the title bar.

Options for Good / Options for Evil (the scales)



Options for Evil, opened with the left button on the scales. The right button opens the same box for Good.

Line	Click
CAN MODIFY LAND ... CAN USE ARMAGEDDON	Each tick box allows or forbids one thing for that side: reshaping land, attacking towns, attacking the leader, and the six powers. Unticked powers do nothing when the icon is clicked.
AGGRESSION slider	Click on the bar between LOW and HIGH to set how aggressively the computer plays that side. Ten positions.
RATE slider	Click between SLOW and FAST to set how often the computer acts.

Game Setup (the globe)



The Game Setup box.

Line	Click
ONE PLAYER / TWO PLAYERS	One player is always in force. Two players (a serial link in the original) is not ported; a notice says so.
PLAY GAME / PAINT MAP	Painting your own map is not ported.
GOOD / EVIL	You always play Good (blue). Playing Evil is not ported.
HUMAN VS AMIGA / AMIGA VS AMIGA	Watching the computer play itself is not ported.
CONQUEST	In a custom game, opens the name field so you can choose a conquest world by name (section 3).
CUSTOM GAME	Turns the current game into a custom game, so the abilities and rules can be changed from now on.
GAME OPTIONS	Opens the Set Game Options box (below). Its OK or CANCEL returns here.
SAVE A GAME	A Mac save dialog. Games are written in the original's file layout, named after the world. Also Cmd-S.
LOAD A GAME	A Mac open dialog for a saved game. Also Cmd-O.
MOVE TO NEXT MAP	Custom game: a fresh random map, sometimes on the next landscape. Conquest: the current world again from the start.
RESTART THIS MAP	The same map again from the beginning.
SURRENDER THIS GAME	Gives up: the world counts as lost and the score box appears.

Set Game Options (the rules of the land)



Each line shows the rule in force; a click switches it to its opposite.

Shown	Click changes it to	Meaning
WATER IS HARMFUL	WATER IS FATAL	Whether a person who walks into the sea is hurt or drowned.
SWAMPS SHALLOW	SWAMPS BOTTOMLESS	Whether a swamp disappears once it has swallowed somebody.
CAN BUILD	CANNOT BUILD	Whether land may be reshaped at all.
BUILD UP AND DOWN	ONLY BUILD UP	Whether the right button may lower land.
BUILD NEAR PEOPLE	BUILD NEAR TOWNS	Where reshaping is allowed: near any of your people, or only near your settlements.

13. Winning, losing and moving on

A world is decided when one side has nobody left, or when you surrender. The game stops and the score box appears.



The score box after a win.

Button	What happens
NEW GAME (after a win, or in any custom game)	After a conquest win the lord congratulates you and names the next world; a click or Return on him opens that world's World to Conquer box. In a custom game a fresh map is made.
TRY IT AGAIN (after a lost conquest world)	The same world again from the start.

The score counts battles won, knights, towns and castles, with a bonus for killing more than you lost. A good score skips worlds: every 5,000 points carries you one battle further than the next one, so the lord may send you several worlds ahead. The title bar shows the result and the next world's name while the box is up, and **Return** moves on.



The lord's speech after a win. His title for you (Mortal, Immortal, Eternal, Deva, Greater Being, Deity, Greater Deity, Mortal God, Greater God, Eternal God) rises with the world reached. Click anywhere or press Return to continue.

14. Keyboard and menu reference

During play

Key	Action
Arrow keys	Scroll the close-up one cell.
Esc	Back to the hand pointer (cancels looking, placing the magnet or placing a swamp).
P	Pause and resume (the zzZ icon).
M	Music on and off.
S	Sound effects on and off.
Return	Once a world is decided, move on to the next one.
[and]	Previous and next conquest world. A convenience of the port, not in the original.
Cmd-N	New Game: opens the World to Conquer box (Esc closes it again).
Cmd-O	Open a saved game.
Cmd-S	Save the game.
Cmd-F	Enter or leave full screen.
Cmd-Q	Quit.

While a box is up

Key	Action
Return	Presses the left-hand button of the box (START GAME, OK, NEW GAME, TRY IT AGAIN, or continue past the lord).
Esc	Closes the World to Conquer box if you opened it with Cmd-N. Has no effect on the other boxes.
Letters, digits, Delete	In the name field: type a world's name, up to fifteen characters.
Any key	Ends the demo.

Menu bar

Populous: Quit. **Game**: New Game, Open Saved Game, Save Game. **Window**: Enter Full Screen. Closing the window quits the game.

15. Messages in the title bar

The original Populous says nothing when a click is refused. The port uses the window's title bar to explain. During play the title normally reads, for example:

```
Populous - GENESIS (battle 0) - mana 1240 - people 14 vs 9 - settle
```

followed by *looking*, *place magnet* or *place swamp* when the pointer is in one of those modes, *music off* and *effects off* when they are, and *ARMAGEDDON* while it runs. A custom game shows CUSTOM GAME instead of a world name. Notices replace the line for a few seconds:

Notice	Why
You can only reshape land while some of your people are in view	You clicked in the close-up with none of your people inside it.
The game is paused - click zzZ or press P to go on	Pause is on.
This world does not allow raising (or lowering) land	The rules of this world forbid it.
Not enough mana to reshape land	You have fewer than 10 mana.
Looking mode - click the up/down arrows icon or press Esc to reshape land again	You clicked empty ground while the pointer was the shield.
These can only be changed in a custom game	You clicked an ability or rule in a conquest world.
Two-player games / Painting the map / Playing as evil / Computer against computer are not ported	You clicked one of those lines in Game Setup.
paused (click zzZ or press P to go on)	Shown steadily while paused.
well done, you conquered ... now battle at ...	The world is won; Return moves on.
... is lost - score ...	The world is lost; Return plays it again.
stopped: ... is not ported yet	The simulation reached a path the port does not have. Start a new world.

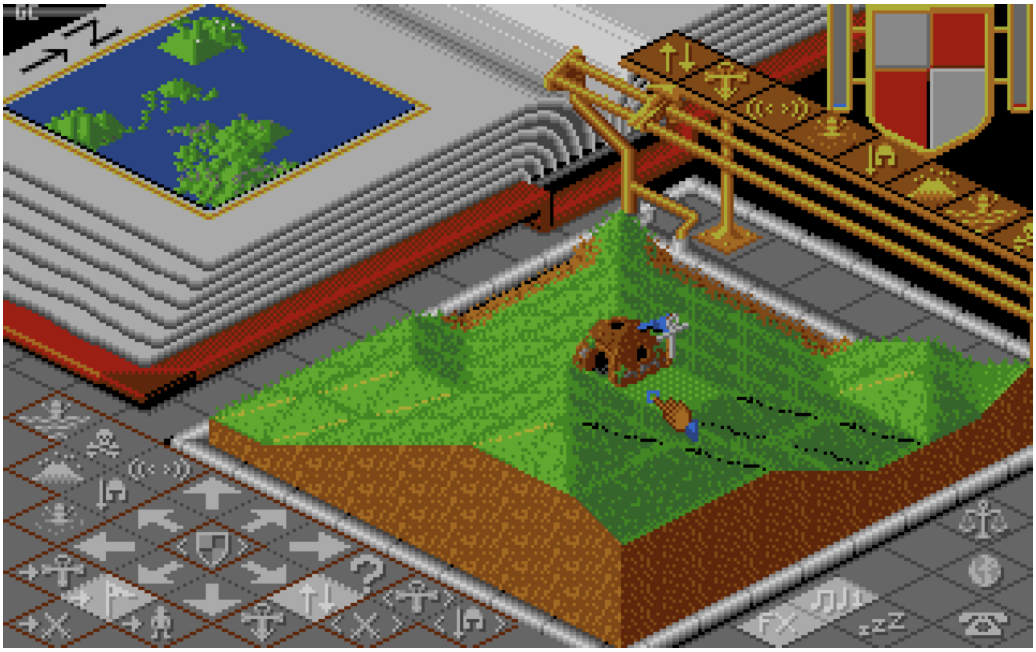
16. What the port does not do

- Two-player games over a serial cable, painting your own map, playing as Evil, and computer against computer. The lines are in the Game Setup box but only put a notice in the title bar.
- The message icon (the telephone) at the lower right of the board.
- The flying creatures that cross the map in the original.
- The original's own marking of lit icons: the port brightens the stone instead.
- The original's speed is not measured; the port runs eight ticks a second.

Everything else on the board, in the boxes and in the simulation follows the original program, which was disassembled and rewritten routine by routine and checked against the original running in an emulator.

A reminder of the whole loop

- Flatten land around your walkers with left and right clicks. Only works with your people in view.
- Keep the behaviour on **Settle** so they build. Watch the population bars and the needle on the rail.
- Use the map and the go-to icons to find the enemy. Scroll the close-up over them and use **Earthquake** or **Volcano**.
- Send everyone to the magnet, make a **Knight** of your leader, then switch to **Fight** when you are stronger.
- When the enemy is gone the score box appears: NEW GAME, listen to the lord, START GAME on the next world.



Genesis a few minutes in: a hut, a walker and the shield following the leader.